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PERTRONIC
AUXILIARY PSU MONITOR - 24V
WITH TIMER
TECHNICAL MANUAL
AUSTRALIA

Valid for:

PCB Hardware:	**v1.03**
PCB Firmware:	**v1.10**

Issue 2.1

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DATE OF MANUFACTURE:

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This document is to be used only for operating Pertronic Auxiliary PSU Monitor 24V, 4A Power Supply Unit.

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Acronyms:

BCO	Bell Cut-Off	Bell Cut-Off switch terminates the output from the panel BELL relay. Two BCO switches are provided, an External BCO switch for Brigade and Tester use and an Internal BCO for Fire Technician use.
DBA	Direct Brigade Alarm	Fire Sprinkler pressure switch/transmitter unit.
EOL	End of Line	End-of-Line termination, nominally 10kΩ Resistor, used to monitor the presence and integrity of the detector circuit.
LED	Light Emitting Diode	
MCP	Manual Call-Point	
OC	Open-circuit	
PCB	Printed Circuit Board	
PSU	Power Supply Unit	
SC	Short-circuit	

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1. INTRODUCTION

The **Pertronic Auxiliary 24V-4A PSU** incorporates a 4 Amp Switch-Mode Power Supply with voltage monitoring and battery test facilities.

Two 12V, up to 38Ah sealed lead acid batteries may be fitted in the **PSU** cabinet.

Up to 10 Amps may be supplied by the batteries and regulator to a load connected to the **PSU**.

A microprocessor monitors the battery voltage and performs timing functions for the 24-Hour Auto test.

Three indicator LEDs are mounted on the door of the **PSU:**

- The green **Mains ON** LED illuminates when the Mains supply is present.
- The yellow **Autotest** LED illuminates when the PSU performs the 24 Hour Auto-Test.
- The yellow **Battery Fault** LED pulses to indicate faults associated with the system.

Change-over relay contacts provide non-latching fault indication to external equipment.

The Fault relay is normally energised when the **PSU** has no fault conditions, and it releases to indicate a fault. Release of the Fault relay is delayed for a period determined by the type of fault.

Operation of the on-board Reset switch (momentary action) initialises the 24 Hour Test Timer. Immediate commencement of the 24-Hour test can also be initiated through the action of the Reset switch.

Figure 1-1: Auxiliary 24V-4A Power Supply Unit – External View

Figure 1-2: Auxiliary 24V-4A Power Supply Unit – Internal View

2. SPECIFICATION

Cabinet:

Dimensions:	310 x 450 x 170	- H x W x D (mm)
Weight:	8.35kg	- excludes batteries.
Material:	1.2mm mild steel, powder-coated beige	

Power Supply:

AC Input	200 – 260Vac, 50/60Hz. Single Phase	
Input Current:	0.8A ac max with 4A dc load.	
Turn-on Time:	120ms at 230Vac 25°C	
Input Protection:	Internal 1A slow blow fuse	
Inrush Current:	Limited by internal thermistor.	
Output Voltage:	27.4Vdc at 20°C	- nominal
Temperature coefficient:	0.145%/°C set for battery charging	
Charger Current:	3A Continuous	
	4A Peak	- 24 hours maximum duration
Peak Output Current (NZ):	10A	- with battery
Battery Type:	2 x 12V, 38Ah Sealed Lead-Acid (GelCell)	
Battery Test:	Automated 60 minute Load Test every 24 Hours	

Controls:

Reset Switch:	Reset / Timer initialisation switch	
Front Panel Indications:	Mains ON LED	- green
	Auto-Test LED	- yellow
	Battery Fault LED	- yellow
Defect Relay:	Normally energised. NO/NC contacts available 1A @ 24Vdc contact rating	
Operating Temperature Range:	0°C to 40°C	
Switch-mode 4A PSU Tested to:	AS/NZ 3100:1997	

Battery Calculator:

A Battery Calculator is available to estimate the battery and charger requirements of a particular system on the Pertronic website:

http://www.pertronic.com.au/battery_calc/welcome.php

Battery Dimensions: Typical – dimensions from Haze Battery

Battery Model	Amp/Hours	Dimensions: W x H x D mm : Kg	Terminals	Max I_{charge}
HZY12-12	12	150 x 97 x 98 (103) : 4.0Kg	D-T2	2.4A
HZY12-18	18	181 x 76 x 167 : 5.9Kg	C-M5	4.5A
HZY12-26	26	166 x 176 x 126 : 8.8Kg	C-M5	6.5A
HZY12-33	33	195 x 130 x 160 : 10.9Kg	B-M6	8A
HZY12-44	44	197 x 165 x 170 : 14.0Kg	C-M6	11A

3. FUNCTIONAL DESCRIPTION

3.1 Auto Test

The **Auxiliary 24V-4A PSU** monitors the battery voltage and operates the power supply at reduced voltage for 1 hour every 24 hours to perform a battery capacity test.

During the 1 hour reduced voltage period, the yellow **Autotest** LED is ON.

Auto test will not be performed if the Mains supply is absent.

3.2 Mains Lost

In AUS mode, a 'Mains Lost' condition will activate the fault relay signal only after the Mains supply voltage has been continuously absent for more than 55 minutes.

In both NZ & AUS mode, the Fault LED signals 'Mains Lost' immediately on loss of mains supply.

The battery presence and 24-Hour tests are not performed if the Mains supply is absent.

3.3 Reset Switch Operation

A momentary action of the **RESET** switch resets and restarts the 24-hour timer. The battery test commences 23 hours after reset, and it then recurs every 24 hours subsequently.

Pushing the **RESET** switch for at least 2 seconds starts the 1-hour battery test immediately, with the **Auto Test** LED energised.

Pushing the **RESET** switch for about half a second cancels the 1-hour battery test. The next battery test will then start 23 hours from this time. This is useful for a quick check to see if the batteries properly connected.

3.4 Australia/New Zealand Selection

In 'AUS Mode' the 'Mains Lost' relay indicates a fault condition only if the Mains has been disconnected for at least 55 minutes. This feature is not supported in 'NZ Mode'

Jumper (J1) determines whether the PSU is in NZ or Aus mode.

NZ Mode	Link **(J1)** absent
AUS Mode	Link **(J1)** fitted

3.5 Fault Relay Operation and Battery Fault Indication

The Fault relay, located at the bottom of the PCB, is normally-energised and de-energises to indicate a Fault has occurred. The delays for the Fault types are shown in the table below.

Four non-latching fault conditions are detected:

The yellow **Battery Fault** LED indicates which Fault conditions are active. The indication is immediate for **High** and **Low Battery/Charger Voltage** and **Mains Lost** faults; but, the indication is delayed for **Battery Capacity** and for **Battery Absent** faults.

Encoded Fault Indications and Fault Relay Delay times			
Fault	**Value**	**Position in Encoded Pulse Stream**	**Delay Before Activating the Fault Relay**
High charger voltage	> 28.8 V	1st	20 seconds
Low Battery (charger in 24 hour test or mains off) Charger low	< 24.4 V	2nd	20 seconds
Battery Capacity Voltage or Battery Absent	< 24.4 V - checked every 20 seconds	3rd	30 - 40 seconds
Mains Lost		4th	Aus: 55 minutes NZ: no fault if battery OK

Table 3-1: Fault Relay Operation and Battery Fault Indication

The **High Charger Voltage** and **Low Battery/charger Voltage** conditions are not tested when the Mains supply is disconnected, or during 24-Hour test.

3.6 Battery Charger Setting

The battery charger voltage is factory set to 27.4Vdc ± 0.05V at 20°C and is temperature compensated. The charger voltage decreases as the ambient temperature rises and increases as the ambient temperature drops.

4. INSTALLATION AND COMMISSIONING INSTRUCTIONS

Each power supply is supplied in a cardboard carton with polystyrene inserts to protect the unit against normal handling during transport. To reduce the possibility of damage the power supply should be stored in its carton until ready for use.

4.1 Mounting

The power supply is designed to be wall-mounted. It should be mounted in a clean dry area, which provides natural convection between the unit and the wall and ensures the cooling slots are not restricted in any way.

4.2 Connections

The PSU is fitted with a termination block with an integral three-pole switch for Mains wiring connection.

The Load Output (**DC OUT**) uses a 2-way terminal block on the bottom of the Monitor Board.

Care must be taken to prevent reverse polarity connections between the power supply and load.

The Fault relay provides clean NC-C-NO contacts rated 1A @ 24V.

4.3 Commissioning

For initial commissioning, it is recommended that the power supply be connected to the Mains without load.

The green **Mains ON** LED should illuminate.

The load should be connected to the power supply and run without the batteries connected.

With the load connected, connect the two 12V batteries to the power supply. Take care to observe the correct polarity.

Allow the batteries to charge in accordance with the manufacturer's recommendation.

Press the **RESET** switch for about 2 seconds. After a few seconds, the **Auto Test** LED will be illuminated indicating that the 1-hour-long battery test has started. Check for any fault conditions, specifically missing battery. Future auto tests will occur at 24-hour intervals since the time of this reset.

Pushing the **RESET** switch briefly cancels the 1-hour battery test, reverting to charging mode, and the next battery test will commence after 23 hours.

Provided the battery has adequate charge and no fault/defect is present, invoke the battery test, and then activate the connected load (such as Bells, amplifiers and the like) to verify it works OK.

Thereafter, cancel the battery test, and repeat the load test, and verify the battery voltage is increased due to the charger being active.

As applicable, ensure the Fault/Defect output contacts are connected to the Fire Alarm 'External Defect/Fault input' to monitor any PSU/battery problems.

5. MAINTENANCE AND REPAIR INSTRUCTIONS

Annual routine maintenance should not be required, provided the power supply unit is installed and operated in accordance with Section of this Technical Manual.

General fault finding work should proceed by checking that the green **Mains ON** LED is illuminated and checking for fault indications encoded in the **Battery Fault** LED.

If a fault or faults occur, carry out the following actions:

Fault	Position in Encoded Pulse Stream (Long Pulse)	Action Required
High Battery/Charger Voltage	1st	Check the current charger voltage. This should be between 26.6Vdc and 28.0Vdc. If it is above 28.0Vdc, replace the Switch Mode PSU. If the charger voltage is within the allowed limits, reset the system and check again for high charger voltage. If the fault persists over subsequent maintenance visits, replace the Switch Mode PSU.
Battery capacity test failure Low battery / charger voltage	2nd	Check the current charger voltage—if necessary, without load or batteries. This should be between 26.6Vdc and 28.0Vdc. If it is below 26.6Vdc, replace the Switch Mode PSU. If the charger voltage is within the allowed limits, reset the system, and check again for low charger voltage. If the fault persists over subsequent maintenance visits, replace the Switch Mode PSU, or batteries.
Battery Absent	3rd	This fault indicates the battery has failed or is not electrically connected to the monitor board. Check the battery leads and connections. If leads and connections are correct, replace the batteries, and check correct operation by starting an Autotest (refer to Reset switch). Note: this fault condition is not tested for during mains failure, and for the 60 minutes during a 24-hour test.
Mains Lost Applies to Australia mode only; feature is not used in New Zealand	4th	This indicates that the Mains supply was disconnected for at least 1 hour since the last maintenance check. Identify the cause for the Mains failure, and take the appropriate action.

Table 5-1: Maintenance and Repair Instructions

6. DIAGRAMS

6.1 System Wiring

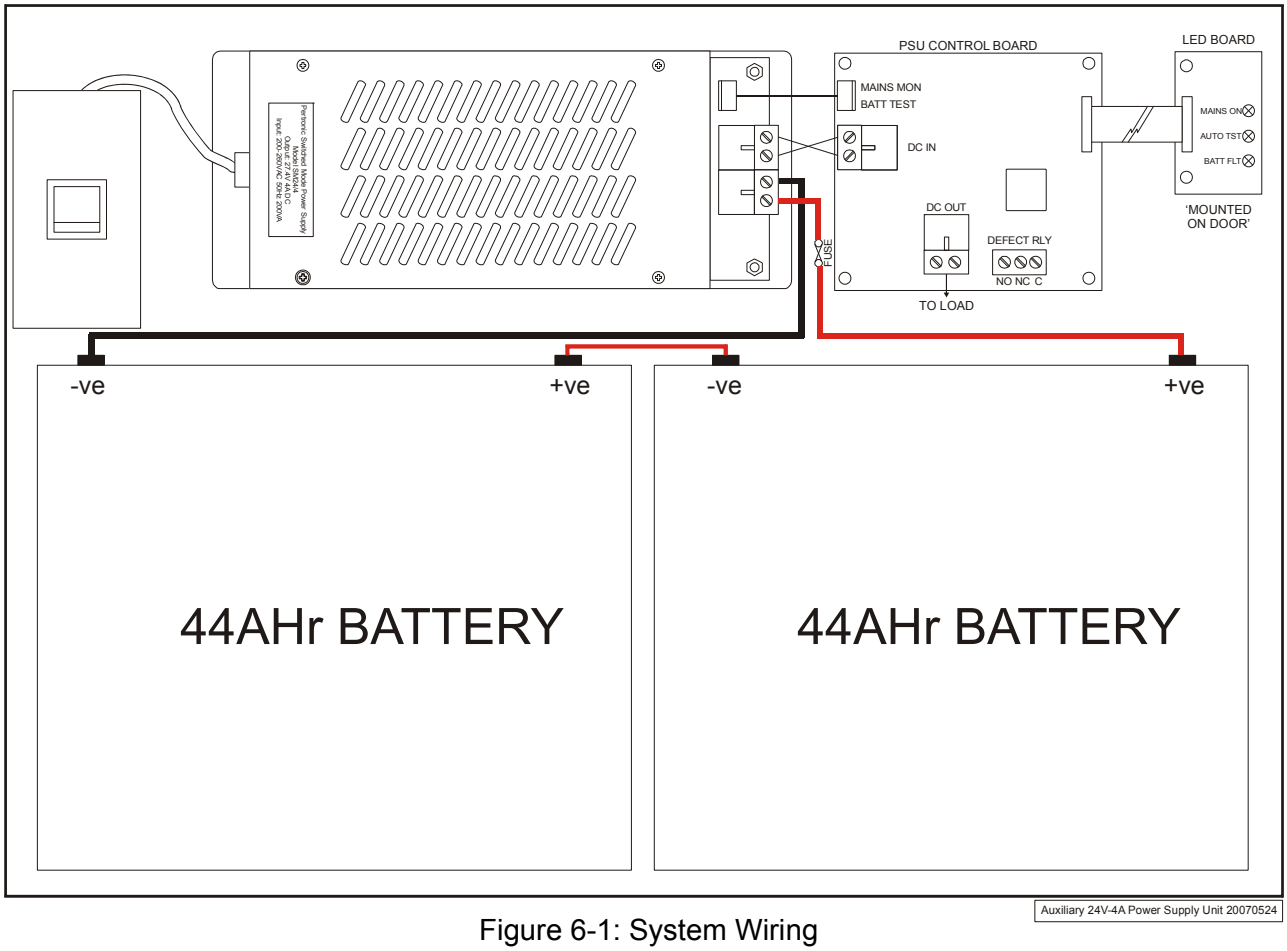

Figure 6-1: System Wiring

6.2 PSU Monitor Board

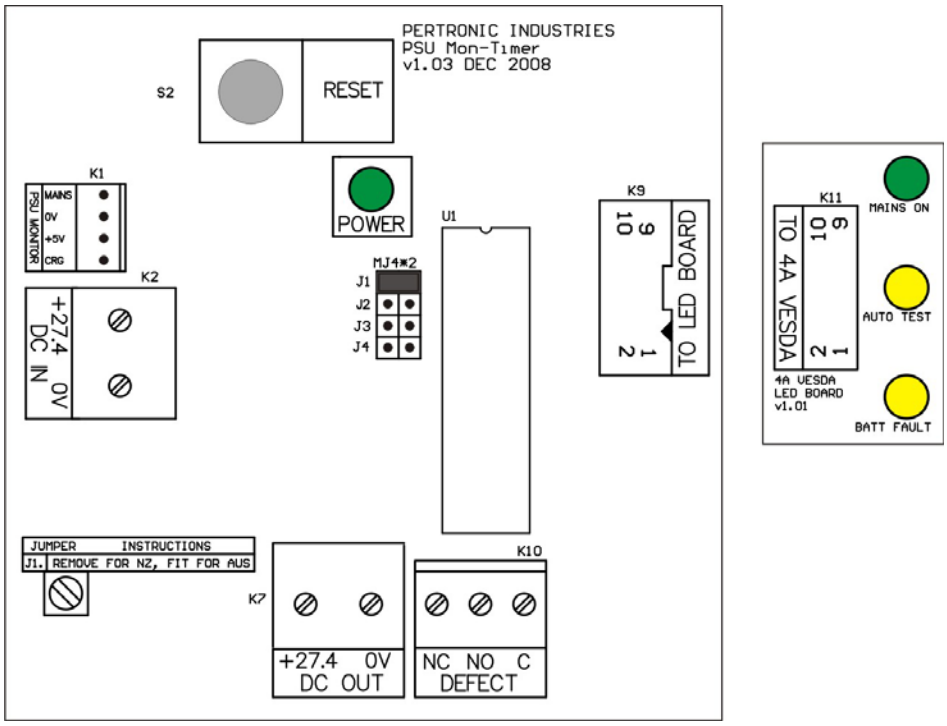

Figure 6-2: PSU Monitor Board and LED Board

7. PRODUCT CODES:

Description	Code
24V, 4A Monitored PSU in BATBOX38	AUX24/4PSU-K003
24V, 4A Switch Mode PSU	24/4SWMPSU
PSU Monitor Board	24VCB
Battery Box: 450 x 315 x 185 (W x H x D mm) - maximum Battery size: 44AHr	BATBOX38

Table 7-1: Product Codes

Document Change History

Issue Number	Date	Reason for Update	Description of Changes	Author
Issue 2.1	April 2013	Created Australian Version		RDB